



Automotive Mobile App for Vehicle Owner Self-Service

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Abstract

Drive Care PRO is a digital vehicle self-service platform for customers, service advisors and administrators. It supports registration/login, vehicle management, service appointments, digital service records, service centers, roadside assistance, notifications and advisor/workshop functions using HTML/CSS/JavaScript, Node.js + Express REST APIs and MongoDB/Mongoose.

Problem Statement & Objectives

Problem: Traditional vehicle service management needs a digital platform.

- Manual booking: calls and physical visits take time.
- Scattered records: history, invoices and warranty details are difficult to manage.
- Emergency gaps: roadside assistance needs a faster digital channel.
- Limited updates: customers need timely service and appointment information.

Objectives:

1. Digitize vehicle and service information.
 2. Provide online booking and status tracking.
 3. Connect users with service centers and roadside assistance.
- Scope: Customer, Advisor/Service Center and Admin modules.

Existing vs Proposed System

Feature	Existing System	Proposed System
Booking	Calls / physical visits	Service appointment booking
Records	Scattered history / invoices	Digital service records
Updates	Limited information	Appointment notifications
Emergency	Separate / manual	Roadside assistance
Gap	Disconnected tasks	ONE APP: user → API → server → DB

Feasibility Study

- **Technical:** Uses HTML/CSS/JavaScript, Node.js, Express.js, REST, Routes, Controllers, Middleware, Mongoose and MongoDB.

Economic: Uses the project's available/open-source web technologies and development tools.

Operational: Centralizes registration, vehicle management, booking, records, notifications and assistance.

Schedule: PBL 2026–27 project for semester implementation and demonstration.

Results & Testing

Key outcomes:

- Authentication and protected routes
- Vehicle management
- Appointment booking and status tracking
- Digital service records
- Service centers and notifications
- Roadside assistance

Testing: CRUD, input validation, protected API routes and REST/database integration.

Conclusion

Drive Care PRO provides a centralized digital experience connecting vehicle owners, service centers and backend services.

It covers authentication, vehicles, appointments, service records, service centers, notifications and roadside assistance.

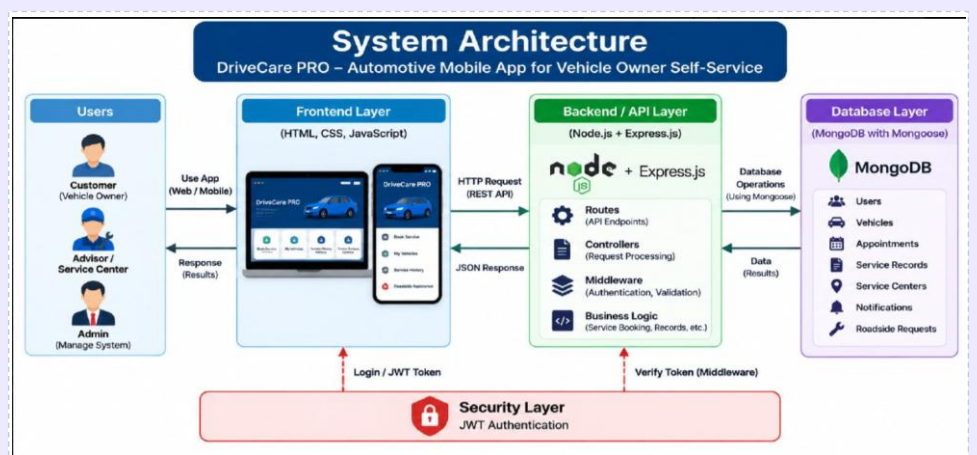
Future Scope

- AI-based maintenance prediction
- Live roadside assistance tracking
- Online payment integration
- Mobile application using Flutter
- Advanced analytics

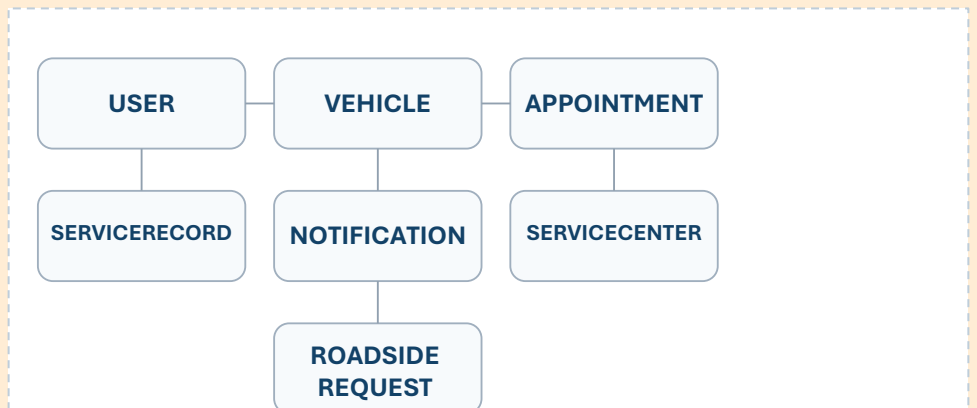
Tools & Technologies

- Front-End: HTML • CSS • JavaScript
- Back-End: Node.js • Express.js • REST APIs
- Business Layer: Routes • Controllers • Middleware
- Database: Mongoose • MongoDB
- Security & Tools: JWT • Postman • GitHub

System Architecture



ER Diagram & Database Design



Implementation Screenshots

